

STREAM NAVIGATOR — HOME MAINTENANCE ENGINE AUDIT REPORT

StreamNavigator — Engine Audit
2026-09-21

Status, 2026-09-20 (same day): all eight Required Changes below have shipped. The two Critical fixes (the silent category default, the thin-result refund), the price cut to \$39, the material/type question, the prior-repair signal, and the design/font cleanup are all live in the repository — see the strikethrough notes under Required Changes for what changed and where. The one piece that could not be completed autonomously: **a new \$39 Stripe Payment Link could not be minted** — no Stripe credentials were available in the environment that made this change, so the checkout button is switched off (``REPLACE_WITH_39_ONE_TIME_LINK``) rather than either charging the old \$59 or pointing at a link that doesn't exist, following the exact pattern already shipped on ``subscriptions.html`` for the identical situation. Creating that link, and deactivating the old \$59 one, requires a few minutes in the Stripe Dashboard — see the setup comment in ``home-maintenance.html`` for the exact steps.

Conducted 2026-09-20 against ``/home-maintenance`` and ``/closing`` on the live site, ``navigator-home-maintenance-engine.js``, ``api/_lib/navigator-engine.js``, ``api/navigator-intake.js``, ``prices.config.json``, and ``tests/`` at the repository's current ``main``. This is a second, broader pass over a product that was already audited once today — ``docs/HOME-MAINTENANCE-AUDIT.md`` (committed ``330217b`/`6b7ca43``) found and fixed a critical defect: a single model call told to invent "typical cost ranges" it had no data for. That fix is live, tested, and confirmed working in this audit. This report does not re-litigate that finding. It applies a wider lens — evidence-scoring discipline, cross-product pricing consistency, the live customer journey end-to-end, and a design comparison against ``/closing`` — and it surfaces findings the narrower audit did not go looking for, including one new critical defect and one confirmed live UI bug.

Access Limitations

Level 1 (direct access) achieved in full. The live page was loaded, scripted, and interacted with — button clicks, form fills, and a real sufficiency-gate rejection were all observed directly in the browser, not inferred from code. ``navigator-home-maintenance-engine.js``, ``api/_lib/navigator-engine.js``, ``api/navigator-intake.js``, and ``prices.config.json`` were read in full. The repository's test suite (86 suites) and the product's own decision harness (``scripts/home-maintenance-audit-harness.js``, 7 scenarios) were both run live this session and passed / produced the outputs quoted below.

What was not done, on purpose: no Stripe payment was completed (per this audit's financial-safety rules) and no paid LLM write-up was generated (the Anthropic account has been out of credit since 2026-09-13 — ``[[no-api-credit-topups]]`` — and the prior audit already established that live report generation is the one thing this product needs a model call for at all). Everything **upstream** of the write-up — the verdict, the sufficiency gate, the pricing wiring, the checkout handoff — is Verified by direct execution, not by reading code and assuming it runs. The **write-up step itself** (does the model, handed the computed verdict, present it without softening or embellishing it) remains Unverified, exactly as the prior audit disclosed, and is not re-litigated here.

Executive Summary

What works: the deterministic core is sound and does what it says. The single Critical defect the earlier audit found — a model inventing "typical cost ranges" from nowhere — is fixed, tested (20 unit tests, all passing), and verified live: submitting the FAQ's own water-heater example through the actual engine on this machine returns the exact verdict the page advertises. The safety override is unconditional and cannot be

reached by cost logic in any test case. The product refuses to manufacture a financial verdict when the customer hasn't supplied the numbers to compute one — a genuine, tested "we will not guess" path that most of this audit's frameworks exist to demand. Nothing on the page claims an insurance benefit, an efficiency saving, or a cost-of-ownership figure it doesn't compute — the prior audit already cut those claims, and they have not crept back.

What is materially weak or broken, found in this pass — all fixed same-day; see Required Changes for what shipped:

1. **A confirmed live UI bug lets a customer pay for the wrong system's

report without ever knowing it. **The category selector shows no default selection on page load — no chip is visually active — but the underlying script silently defaults to "Roof" regardless. A customer deciding about a water heater or a generator who fills in every other field and never happens to click a category chip will submit, and be charged \$59 for, a Roof analysis. This cannot be caught by the sufficiency gate, because "Roof" is a valid category value.** Verified live** on this session's fresh page load.

1. **No thin-result safeguard exists for the one outcome this product

explicitly, deliberately sells: a \$59 report with no financial verdict in it.** A customer can check "I don't have a quote for either yet" — an answer the product calls "honestly scoped, not blocked" in its own test suite — pay \$59, and receive nothing but a boilerplate age-range sentence and a static, category-level checklist. Two sibling products in this same codebase (Government Money Finder, Contractor Navigator) already auto-refund exactly this shape of outcome. Home Maintenance has no such trigger anywhere in its code.

1. **The price sits above two mechanically more complex siblings.** Contractor

Navigator (\$49) extracts real uploaded documents and checks them against a held reference database of statutory caps and published cost ranges — more real work, for less money. Government Money Finder (\$39) runs a larger deterministic rules engine against a bigger reference table, for a third less money. Home Maintenance's engine (285 lines) is the smallest and simplest of the four self-serve deterministic engines in this product line, has no document-extraction step, holds no external reference-cost database, and confers no recurring benefit — yet costs more than two products that do more.

Biggest customer-value problem: #2 above. A product whose own honesty discipline is its main selling point (never guess a cost) currently has no mechanism to stop itself from charging full price for an answer that isn't one.

Biggest product opportunity: the product is honestly scoped as a single-decision repair-vs-replace calculator, not the broader "which maintenance is worth doing, and when, across my whole house" advisor this audit's own brief assumes as the category's ideal. That narrower scope is not a truthfulness problem — the page never claims to do more — but it is a real ceiling on customer value. See Recommended Product Specification.

Is the current price justified? No. **Recommended price: \$39, one-time** (down from \$59) — see Pricing Audit.

Promise vs. Delivery

Promise	What the Product Actually Does	Gap	Severity	Recommended Fix
"Get a clear repair-vs-replace recommendation grounded in real numbers" (H1)	Verified: <code>`analyze()`</code> computes the ratio purely from the customer's own <code>`repair_quote`/`replacement_quote`</code> ; no dollar figure is invented anywhere in the engine or the write-up prompt	None — this is the one promise the prior audit's fix was built to keep, and it holds	—	None needed
"What you get": a repair-or-replace call "using your own quoted numbers"	True when both quotes are given. When neither is given, the customer instead gets a boilerplate age sentence and a static checklist — still charged \$59	The bullet is true on average, false for a disclosed-but-real subset of paying customers	High	Auto-refund the <code>`NEED_BOTH_QUOTES`</code> outcome (see Required Changes #1)
"Roof, HVAC, water heater, windows, generator — tell StreamNavigator what you're deciding on" (implies the category you pick is the category you get)	Verified live: no category chip is visually active on page load; the script defaults to "Roof" silently if none is clicked	A customer can be charged for the wrong system's analysis without any indication of the mistake	Critical	Remove the silent default (see Required Changes #2)
"One price. One repair-or-replace decision." / \$59 one-time	Verified end-to-end: page copy, <code>`prices.config.json`</code> (<code>`5900`</code> cents), the live Stripe Payment Link (<code>`buy.stripe.com/28EeVd9ko3I80FS1BuabK09`</code>), and the webhook all agree — a genuine one-time charge, no subscription anywhere in the wiring	None on mechanism. The amount is out of line with sibling products doing more work for less (see Pricing Audit)	High	Cut to \$39, one-time
"Where your system's age sits against a typical service-life range for its category"	True, but the range assumes one material per category (e.g. asphalt shingle for Roof) and the intake never asks which material the customer actually has, even though the engine's own caveat names material as the dominant variable	The one non-customer-supplied fact in the report is disclosed as approximate but could be sharpened cheaply	Medium	Add a one-chip material/type question for Roof and Water Heater
Implicit: the product is the comprehensive "what maintenance is worth doing, house-wide" advisor this audit's brief assumes	Explicitly, honestly narrower: one system, one decision, per \$59 report — the page never claims otherwise	Not a truthfulness gap (the page doesn't overclaim), but a ceiling on total addressable value	— (opportunity, not a defect)	See Recommended Product Specification
Visual/brand promise: "another product in the same Stream Navigator family" as <code>`<closing`</code>	Close but not identical — same fonts, ink/paper palette, and flat pricing-card treatment; still carries a colored logo-mark icon and a pill sub-brand badge <code>`<closing`</code> doesn't have, plus two unused font families loaded for no reason	Cosmetic, already flagged and deliberately deferred on the sibling <code>`<property-tax`</code> page in <code>`docs/PROPERTY-TAX-AUDIT.md`</code>	Low	Drop the logo mark/pill badge; remove the dead <code>Inter/Sora`<link>`</code> tags

Customer Journey Audit

1. Landing page. The hero states the problem ("Repair it or replace it?"), the mechanism ("tell StreamNavigator what you're deciding on... grounded in real numbers"), and proves it with a worked example (14-year-old water heater, \$1,400 vs. \$2,600, badge "Replace") before asking for anything. This is a genuinely strong pattern: the visitor sees the actual mechanism running on real numbers before being asked to trust it. Understandable in well under 30 seconds. **Verified live.**

2–3. Inputs / property information. Structured chips and number fields — category, age (or "not sure"), what's prompting the decision, repair quote, replacement quote (or "I don't have one yet"), optional notes, optional file upload, email. Every field earns its place: category and symptoms gate the analysis path, age drives the lifespan comparison, the quotes drive the only financial claim the product makes. Nothing asks for information the product doesn't use. **One defect found here: the category chip has no visible default and silently resolves to "Roof" if skipped — see Executive Summary #1. Verified live.**

4–5. Maintenance analysis / prioritization. Not applicable in the sense the audit brief assumes (there is no multi-item priority list — this product analyzes exactly one system per report), and the page never claims otherwise. Within its scope, there is a real, tested priority order: safety concern > cost comparison > "we need more information." That ordering is correct and unconditional (see Decision Engine Audit).

6. Cost analysis. Arithmetic only, on the customer's own two numbers, against a disclosed 50% threshold. No invented cost anywhere. **Verified** via direct engine execution (harness output below).

7–8. Results / recommendation. The write-up prompt (``api/_lib/navigator-engine.js:459-471``) is tightly bound to the computed verdict — it is explicitly forbidden from stating a dollar figure that isn't the customer's own, from softening a safety verdict, or from filling a missing-quote gap with generic advice "dressed up as a recommendation." **Unverified in a live paid run** (no model credit available this session), but the prompt itself, read directly, leaves very little room for drift.

9. Next action. Every verdict path ends at a concrete instruction: call a professional now (safety), get one more quote (missing-quote states), or here's your answer plus a contractor question checklist (both-quotes states). No verdict path ends in vague advice.

10. Pricing/payment. Verified live, end to end, this session: filling the real form with an intentionally incomplete submission produced the exact inline rejection the engine's ``checkSufficiency()`` computes ("Enter the system's age, or mark it unknown.") without navigating away — confirming the client-side gate actually runs, not just that it exists in source. The page's own error-handling detail (an inline message that survives a reload, rather than a toast that clears after 3.6 seconds — see ``home-maintenance.html:216-220``) is a real, deliberate usability fix, and it worked as designed.

Maintenance Decision Engine Audit

Verdict logic (``navigator-home-maintenance-engine.js:186-237``), confirmed by direct execution this session:

``` the FAQ's own example — 14yo water heater, \$1,400 vs \$2,600 verdict: replace reason: The repair

quote (\$1,400) is 54% of the replacement quote (\$2,600) — at or above the 50% threshold where replacing is usually the better value. age: 14 yrs vs typical 8-12 (past range: true)

safety concern overrides a dramatically cheap repair (\$200 repair vs \$8,000 replace) verdict: urgent\_safety  
cost comparison: null <- cost math never ran

no quotes at all, age unknown verdict: need\_both\_quotes reason: No repair or replacement quote was provided, so no financial recommendation can be made yet. ``

### Strong decision rules:

- **Safety always wins, unconditionally.** `applySafetyOverride()` runs

first and returns a terminal result; the cost-comparison branch is structurally unreachable once `safety_concern` is flagged, confirmed by a test that deliberately makes repair look dramatically cheaper (\$200 vs. \$8,000) and shows the verdict is still `urgent_safety` with `costComparison: null`. This is exactly the evidence-threshold behavior Section 12 of this audit's own framework calls for: an irreversible, safety-relevant outcome gets acted on regardless of how weak the underlying evidence is (a self-reported symptom checkbox), because the cost of a false positive (an unnecessary service call) is trivial next to the cost of a false negative.

- **The repair-vs-replace threshold is disclosed, not hidden.** Every verdict

states the exact ratio and the exact threshold in plain language ("\$1,400 is 54% of \$2,600 — at or above the 50% threshold"). A customer can check the engine's own math against a calculator in ten seconds.

- **A missing quote never gets filled with a guess.** Three distinct verdicts

(`NEED_REPAIR_QUOTE`, `NEED_REPLACEMENT_QUOTE`, `NEED_BOTH_QUOTES`) exist for exactly this, and the write-up prompt is explicitly forbidden from papering over them with generic advice.

### Weak or missing decision rules:

- **The lifespan-range table doesn't ask the one variable it names as the**

reason it might be wrong. `LIFESPAN_RANGES['Roof']` assumes asphalt shingle and says so in its own caveat; `Water Heater` assumes a storage-tank unit and says tankless units last 15-20 years instead of 8-12 — nearly double. The intake never asks which the customer has. The caveat is honest, but the product could simply ask and stop needing the caveat to do so much work. Evidence Strength for this table's applicability to a specific customer: 3/5\*\* (Moderate — a well-established general range, but a material variable that would change the answer is known to the engine and not collected).

- **The 50% threshold is a single, un-nuanced heuristic.** It does not

account for financing cost, energy-efficiency savings from replacing, or warranty differences — and, correctly, the page no longer claims it does (the prior audit removed the "efficiency and cost-of-ownership differences" bullet rather than fake it). This is an honest limitation, not a truthfulness violation, but it caps how sophisticated a "real economics" claim can be.

- **No repair-vs-replace nuance for a system that has already failed once**

before.\*\* The intake never asks "has this been repaired before," which is one of the single strongest real-world signals for "replace" that a customer could supply for free. This is a missed low-cost, high-value input.

**Incorrect or dangerous assumptions:** none found. The one thing this file's own header calls out as a considered exception (using service-life years, never cost, as context) is followed consistently in the code and in the write-up prompt's hard rules; it does not blur into a cost claim anywhere tested.

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## Savings Engine Audit

There is no "savings" claim on this page at all, and that restraint is itself a finding worth stating plainly: the product does not compute or display an estimated dollar savings figure anywhere, unlike several sibling products. **This is correct, not a gap.** A believable savings number for "which of two customer-supplied quotes is cheaper" would just be ``abs(repairQuote - replacementQuote)`` — trivial, and stating it as "your savings" would imply the product found something, when the customer already had both numbers in hand. The product's actual value is not a savings calculation; it is (a) a disclosed decision rule applied consistently, (b) service-life context, (c) a safety triage step, and (d) a contractor question checklist. All four are delivered when the inputs support them.

**Where the real economic value is, and where it is not:** for a customer who already has both quotes, the marginal value StreamNavigator adds beyond what a free web search for "50% rule repair or replace" would tell them in under a minute is genuinely modest — consistent application of a known threshold, service-life framing, and a checklist. That value is real but should be priced like what it is (see Pricing Audit), not like a document audit that catches errors a customer could not have found alone.

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## Maintenance Schedule Audit

**Not applicable in the form the audit brief assumes,** and this needs to be stated plainly rather than scored as a defect: the product does not produce a maintenance calendar, does not track multiple systems at once, and does not tell a homeowner what to do next season. It produces one repair-or-replace call for one system, once, per \$59. The FAQ says this outright: "Each report covers one repair-or-replace decision... Deciding on more than one system at once? Submit a separate report for each." That is an honest scope statement, not a broken promise — but it means the product does not deliver the "when should this happen, across my whole house" planning value this audit's own framework treats as the category's ideal. See Recommended Product Specification for what closing that gap would require.

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## Risk Audit

**Safety risks:** handled correctly and unconditionally — see Decision Engine Audit. This is the one place where the product explicitly refuses to let cost logic touch the outcome, and it is tested against the least forgiving case (repair dramatically cheaper) to prove it.

**Property-damage / major-repair risks:** the "already past typical service life" caution (``navigator-home-maintenance-engine.js:217-221``) is a real, if modest, early-warning signal — it tells a customer who chose "repair" that the decision is likely to recur soon. It does not, however, warn a customer who chooses **replace** about anything (there's nothing to caution there), and it does not warn about consequential damage risk from delay (e.g., a leaking water heater causing floor damage) beyond the symptom checkbox itself.

**Insurance-related risk:** correctly, entirely absent. Neither the page nor the engine claims an insurance-premium benefit, an insurability benefit, or any interaction with a policy. This is the right restraint — this audit's own brief explicitly warns against manufacturing a premium-savings benefit where the evidence doesn't support one, and this product simply doesn't try.

**Unnecessary-spending risk (false urgency):** none found. The product's entire design is oriented against manufactured urgency — it will not recommend replacement based on age alone (the lifespan table is "context... never a claim about this specific unit's remaining life," per the write-up prompt's own hard rule), and it explicitly refuses to render any financial verdict without the customer's own numbers.

## Evidence Audit

| Claim / Recommendation                   | Evidence                          | Evidence Type                                         | Recency                                   | Evidence Strength (0–5) | Completeness (0–5)                                                           | Decision Confidence (0–5)                             | Key Uncertainty                                                                                                         |
|------------------------------------------|-----------------------------------|-------------------------------------------------------|-------------------------------------------|-------------------------|------------------------------------------------------------------------------|-------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| REPLACE/REPAIR verdict via the 50% ratio | Customer's own two typed quotes   | Customer-Specific + Industry (disclosed heuristic)    | Current                                   | 5                       | 5 (when both quotes given)                                                   | 4                                                     | The 50% rule itself ignores financing, efficiency, and warranty differences — disclosed as a simplification, not hidden |
| Age vs. typical service-life range       | Curated 5-row table, by category  | Industry / Generic                                    | Historical (slow-moving engineering fact) | 3                       | 2 (material/type never asked, despite being the caveat's own named variable) | n/a (context only, never a standalone recommendation) | Roof/Water Heater ranges can be nearly 2x off if material differs from the assumed default                              |
| Contractor question checklist            | Fixed reference list per category | Generic (by design — a reference list, not a finding) | Durable                                   | n/a                     | n/a                                                                          | n/a                                                   | Correctly presented by the write-up prompt as "practical next steps," never as something "discovered"                   |

| Claim / Recommendation | Evidence | Evidence Type | Recency | Evidence Strength (0–5) | Completeness (0–5) | Decision Confidence (0–5) | Key Uncertainty |
|------------------------|----------|---------------|---------|-------------------------|--------------------|---------------------------|-----------------|
|------------------------|----------|---------------|---------|-------------------------|--------------------|---------------------------|-----------------|

|                                     |                                |                   |         |                                       |                                             |   |                                                                                                                                                                                             |
|-------------------------------------|--------------------------------|-------------------|---------|---------------------------------------|---------------------------------------------|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Safety override!<br>urgent guidance | Self-reported symptom checkbox | Customer-Specific | Current | 2 (a checkbox, not a verified hazard) | 5 (sufficient for the *action* it triggers) | 5 | Correctly weighted: low evidence strength is appropriate here because the recommended action (call a professional) is cheap and reversible relative to the downside of a missed real hazard |
|-------------------------------------|--------------------------------|-------------------|---------|---------------------------------------|---------------------------------------------|---|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                                                   |                           |     |         |                                        |                                                 |     |                                                                                                          |
|---------------------------------------------------|---------------------------|-----|---------|----------------------------------------|-------------------------------------------------|-----|----------------------------------------------------------------------------------------------------------|
| "No financial verdict yet" (missing-quote states) | Absence of customer input | n/a | Current | n/a (a "no action" state, not a claim) | 5 (correctly identifies exactly what's missing) | n/a | Does not yet tell the customer *how* to get the missing figure quickly — a copy gap, not an evidence gap |
|---------------------------------------------------|---------------------------|-----|---------|----------------------------------------|-------------------------------------------------|-----|----------------------------------------------------------------------------------------------------------|

|                                                             |                          |             |         |   |   |   |                                                                                                       |
|-------------------------------------------------------------|--------------------------|-------------|---------|---|---|---|-------------------------------------------------------------------------------------------------------|
| \$59 price reflecting "real, computed, decision-grade work" | Cross-product comparison | Comparative | Current | 4 | 5 | 4 | See Pricing Audit — the comparison is strong; the conclusion is a pricing judgment, not a factual one |
|-------------------------------------------------------------|--------------------------|-------------|---------|---|---|---|-------------------------------------------------------------------------------------------------------|

**Audit of the evidence system itself:** the engine does not display a formal 0-5 evidence score to the customer, and it does not need to — its actual behavior already satisfies the substance of Sections 6, 11, 12, and 14 of this audit's framework without the scaffolding. It shows its work (the ``reasons`` array states the exact numbers and threshold used), it never lets a low-cost, reversible caution require strong evidence (the safety override triggers on a self-reported checkbox alone, correctly), and it never lets a "no answer yet" state get filled with a manufactured one. The one place it falls short of its own standard is the lifespan table: it states a caveat about material/type variance without ever asking the one question that would resolve it, which is a data-collection gap, not a disclosure gap.

## Actionability Audit

Every verdict path ends at a specific next action:

| Verdict                                                                  | What the customer is told to do next                                                                        |
|--------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| <code>`urgent_safety`</code>                                             | Contact a licensed professional now; specific gas-leak instructions (leave the property, call from outside) |
| <code>`replace`</code>                                                   | Replace, with the checklist of contractor questions for that category                                       |
| <code>`repair`</code>                                                    | Repair, with the checklist, plus a caution if the system is already past its typical range                  |
| <code>`need_repair_quote`</code> / <code>`need_replacement_quote`</code> | Get the one missing quote — but does not say how quickly, cheaply, or where to get it, a real if minor gap  |
| <code>`need_both_quotes`</code>                                          | Get both quotes before a financial verdict is possible — same gap                                           |

DIY-vs-professional guidance is present only implicitly, through the product does not distinguish tasks a homeowner could reasonably do from those that require a licensed professional, beyond the safety-override case. For the heater, windows, generator), that is defensible — these are not ty means the product offers no DIY/professional judgment at all, wh capability.

## UX Audit (vs. ``/closing``)

**Verified live**, both pages loaded and screenshotted this session. Home Maintenance is visibly the *\*closest\** of the non-`closing` Navigator pages to the `closing` design language — same ink/paper palette, same Newsreader/IBM Plex fonts, same flat card and button treatment on the primary pricing card. This was a deliberate reskin (`navigator-closing-theme.css` is loaded on top of the shared stylesheet), and it mostly works.

**Concrete, confirmed discrepancies:**

- **Header treatment diverges.** Home Maintenance still shows a colored

logo-mark icon box and a pill-shaped "Home Maintenance Navigator" sub-brand badge next to the wordmark. `closing`'s header is plain text: `StreamNavigator / Closing Disclosure Audit`. This exact gap was already identified and deliberately left open on `property-tax` in `docs/PROPERTY-TAX-AUDIT.md` as a known, low-priority cosmetic choice — it applies identically here, uncorrected.

- **Two unused font families load on every page view.** `home-maintenance.html`

loads Inter and Sora (the shared theme's fonts) *\*and\** Newsreader/IBM Plex (the closing theme's fonts) — four families in one page load, when only the second pair is actually used once `navigator-closing-theme.css` overrides the shared rules. This is dead network weight with no visual benefit, on every visit.

- **Smaller components still carry the old system's visual habits** — chip

toggles, check-card ticks, and step-number circles keep rounder corners and small offset shadows that `closing`'s hand-written CSS never uses anywhere (flat 1px hairline borders throughout). Minor, but visible on close inspection.

**Mobile:** verified at 375×812 — the layout holds up well, text remains legible, the header wraps cleanly to two rows, and nothing overlaps or clips. No mobile-specific defect found.

**Verdict:** closer to `closing` than any other Navigator page, but not a match, and the gap is entirely cosmetic residue from the override-layer approach rather than a rebuild — the same story property-tax's own audit already told about itself.

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## Pricing Audit

**Current Price:** ~~\$59~~ **\$39, one-time — shipped.** The page, the button placeholder, and `prices.config.json` all agree on \$39; only the live Stripe Payment Link remains to be created (no credentials available to do it autonomously — see Status note at the top of this report).

**Recommended Price: \$39, one-time.**

**Recommended Pricing Model:** one-time (unchanged) — this is a single, non-recurring decision. It is not in `ENTITLED\_PRODUCTS` (only `rental` and `landlord` carry a year of re-runs), and there is nothing to monitor or re-check after the verdict is delivered. A subscription would sell an ongoing service this product does not render.

**Reasoning.** The house standard, applied twice already in this codebase (`docs/PROPERTY-TAX-AUDIT.md`, `docs/SUBSCRIPTIONS-AUDIT.md`) is to price by mechanism shape and real work done, not by the dollar stakes of the decision the customer happens to be facing. Applying that same standard here,

consistently, against every deterministic-engine product in the line:

| Product                    | Price     | Engine complexity (lines)          | Requires a document?                  | Reference data beyond the customer's own input?   | Recurring benefit?                                                                                                                                                                       |
|----------------------------|-----------|------------------------------------|---------------------------------------|---------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Government Money Finder    | \$39      | 822 (+ 377-line program table)     | No                                    | Yes — a held program-eligibility table            | No                                                                                                                                                                                       |
| Contractor Navigator       | \$49      | 2,163 (+ 373-line reference table) | Yes — an actual contractor estimate   | Yes — statutory caps, published cost ranges       | No                                                                                                                                                                                       |
| Home Maintenance Navigator | \$59      | 285 (smallest in the line)         | No                                    | A 5-row lifespan table only — no cost data at all | No                                                                                                                                                                                       |
| Property Tax Navigator     | \$79      | 329                                | No                                    | No — customer-supplied figures only               | Yes — an appeal saves money every year until reassessment (this is "/property-tax`s own audit's stated reason for costing more than its "mechanically identical twin," Home Maintenance) |
| Insurance / HOA Navigator  | \$79 each | —                                  | Yes — real uploaded documents, diffed | —                                                 | No                                                                                                                                                                                       |

Home Maintenance has the s  
no document, holds no cost-  
Audit), and confers no recurr  
Property Tax costing \*more\*  
Home Maintenance should c  
closest mechanical twin by e  
substantially more real, chec  
\$10 less than Home Mainten

This was not caught by the e  
the cross-product mechanism  
standard — it confirmed the s  
back up) as Insurance and G

products' engine complexity.

### Customer Value Threshold

product's marginal value over commonly-cited personal-finance life framing, and a checklist, than a product that extracts a

## Required Changes

\*(All eight items below shipped in document for the one piece – credentials.)\*

### Critical — Must Fix

\*(None remaining — both sh

1. ~~Add a thin-result auto-refund for `NEED\_BOTH\_QUOTES`.~~ **Shipped.**

`api/\_lib/navigator-engine.js` now sets `refund\_state: 'due\_thin\_result'` the moment `homeMaintenanceAnalysis.verdict === Verdict.NEED\_BOTH\_QUOTES`, picked up by `api/process-refunds.js`'s existing `REFUND\_STATE\_THIN` queue — the same mechanism Government Money Finder and Contractor Navigator already use for the identical shape of problem. The page's field hint and FAQ both now disclose this before payment. Covered by `tests/home-maintenance-page.test.js`.

1. ~~Fix the silent category default.~~ **Shipped.** `home-maintenance.html`'s

inline script now starts `selectedCategory` at `null`; a customer who never clicks a category chip is blocked by `checkSufficiency()` instead of silently defaulting to Roof. Verified live in this session (no chip shows active on load) and pinned by `tests/home-maintenance-page.test.js`.

## High Priority

\*(Both shipped; one is code-complete pending a Stripe credential.)\*

1. ~~Cut the price to \$39, one-time.~~ **\*\*Code-complete, Stripe link**

pending.\*\* The page displays \$39, `prices.config.json` moved this page to `\_skipped` with the reasoning recorded, and the checkout button is a switched-off `REPLACE\_WITH\_39\_ONE\_TIME\_LINK` placeholder rather than either charging \$59 or pointing at a dead link — exactly the `subscriptions.html` pattern, verified live: the button correctly refuses to navigate and shows a clear message instead. Creating the new Stripe Price/Payment Link and deactivating the old \$59 one needs a human with Stripe Dashboard access; the setup comment in `home-maintenance.html` gives the exact steps.

1. ~~Ask the material/type question for Roof and Water Heater~~ — **Shipped.**

`navigator-home-maintenance-engine.js` now holds a `MATERIAL\_RANGES` table for both categories (metal/tile/slate roofing, tankless water heaters), additive to the existing default table so every prior test still passes unchanged. The page renders the material chips only for these two categories, reading the options

from the engine so the two can't drift apart. Covered by new tests in ``tests/home-maintenance-engine.test.js``.

## Medium Priority

\*(Both shipped.)\*

1. ~~Tell the customer how to get a missing quote quickly~~ — **Shipped.**

The write-up prompt (``api/_lib/navigator-engine.js``) now instructs the model to point a missing-quote customer at the checklist's own "get another quote" reminder as the concrete next step, rather than only naming what's absent.

1. ~~Ask whether the system has already been repaired before for the same

issue.~~ **Shipped.** A new checkbox feeds ``already_repaired_before`` into the engine, which adds a disclosed caution on a ``repair`` verdict without ever changing the verdict itself — additive, exactly like the existing age-based caution. Covered by new tests in ``tests/home-maintenance-engine.test.js``.

## Low Priority

\*(Both shipped.)\*

1. ~~Drop the colored logo-mark icon and pill sub-brand badge from the

header~~ — **Shipped**, on ``home-maintenance.html`` (the ``/property-tax`` instance of this same gap is unchanged and remains open there).

1. ~~Remove the unused Inter/Sora ``<link>`` tags~~ — **Shipped.**

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## Recommended Product Specification

**What Home Maintenance Navigator does today, correctly, and should keep doing:** one repair-or-replace decision, arithmetic only on the customer's own two quotes, an unconditional safety override, a disclosed threshold, and a refusal to guess a cost it doesn't have. Do not weaken any of this to build the expansion below — it is the product's actual credibility.

**The natural next-tier product, if StreamNavigator wants to capture the fuller "which maintenance is worth doing, house-wide, and when" value this audit's brief describes as the category's ideal,** without touching the existing \$39 single-decision product:

- **Inputs:** a one-time system inventory per property — category + age for

each major system the homeowner has (roof, HVAC, water heater, windows, generator, plus water/electrical/plumbing basics) — reusing the exact ``CATEGORIES`/`LIFESPAN_RANGES`` vocabulary already built for this product.

- **Calculation:** for each system, where it sits against its typical

service-life range (same deterministic table, no new cost claims) — producing a **prioritized list by proximity to end-of-typical-life**, not by invented dollar risk. Systems past their range are flagged for budget

planning; systems well within range are explicitly told "nothing to do yet," consistent with this audit's Section 20 standard that a genuine "do nothing" answer is a valuable, not a wasted, output.

- **Output:** a household maintenance-timing calendar — what to start

budgeting for and roughly when — explicitly **not** a cost estimate (the same discipline this product already holds), paired with an offer to run the existing \$39 repair-vs-replace report the moment any specific system actually needs a decision.

- **Pricing:** this is a materially larger, recurring-relevant product (a

household's system inventory is worth re-checking annually) and could reasonably justify an annual price along the lines of Insurance/HOA (\$79), or a modest add-on to an existing Navigator bundle — but this is a new product decision requiring its own audit before a number is set, not an extension of today's \$39 recommendation.

This is explicitly a **future opportunity**, not a required fix — the current product is honest about not doing this, and should not be marked down for a scope it never promised.

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## Final Customer-Value Test

*\*\*If I were a rational homeowner paying the recommended price, would I reasonably expect this product to save me more money than it costs by helping me spend money on the right maintenance at the right time while avoiding unnecessary maintenance spending and preventable expensive problems?\*\**

### Customer Value Assessment

**At \$39, yes, for the customer who has both quotes in hand** — which the product's own field hint tells them to get before paying, and which is the condition under which the engine delivers its actual, computed, non-generic answer. The disclosed 50% threshold, applied consistently and shown in the customer's own numbers, is worth more than the ten seconds it takes to compute, because it comes bundled with the service-life context, the safety triage, and the negotiation checklist — genuine, if modest, value beyond doing the division yourself.

**At \$59 today, and for the customer without quotes yet, no** — not until Critical fix #1 ships. A customer who honestly discloses they have no quotes, exactly as the product invites them to, currently pays full price for a category checklist and a boilerplate sentence. That customer should either not have been charged, or should get their money back automatically, the way two sibling products in this same codebase already handle the identical situation.

### Recommended Product

Keep the deterministic engine, the safety override, and the refusal to invent a cost exactly as they are. Fix the silent category default. Add the missing thin-result refund. Ask one more question (material/type) to sharpen the one piece of non-customer-supplied content in the report.

### Recommended Price

**\$39, one-time.**

## **Highest-Impact Fix**

**The thin-result refund (Critical #1).** Every other issue in this report is a matter of degree — a price that's somewhat high, a design detail that's somewhat inconsistent, a question that could sharpen a range. This one is categorical: it is the difference between a product that keeps its core promise ("never a guess at what either one usually costs — and never a charge without an answer") and one that quietly breaks it for a disclosed, tested, intentionally-supported subset of its own paying customers.